

Differentiability

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def: A function $f: X \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ is called differentiable at some point $\vec{a}$ in the interior of X (i.e. there is $\delta > 0$ such that $B_\delta(\vec{a}) \subseteq X$) if there is a linear function $h: \mathbb{R}^n \rightarrow \mathbb{R}$

given by $h(\vec{x}) = \vec{m} \cdot (\vec{x} - \vec{a}) + f(\vec{a})$

such that $\lim_{\vec{x} \rightarrow \vec{a}} \frac{f(\vec{x}) - h(\vec{x})}{\|\vec{x} - \vec{a}\|} = 0$.

def: We say $f: X \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ (where the domain X is an open set in $\mathbb{R}^n$) is differentiable if f is differentiable at every $\vec{a}$ in X

Theorem: If f is differentiable at $\vec{a}$, then f is continuous at $\vec{a}$.

Theorem: $f: X \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ is differentiable at $\vec{a}$ if and only if all partial derivatives of f exist at $\vec{a}$ and

$$h(\vec{x}) = f(\vec{a}) + \vec{\nabla} f|_{\vec{a}} \cdot (\vec{x} - \vec{a}) \quad (\text{i.e. } \vec{m} = \vec{\nabla} f(\vec{a}))$$

$$\text{ satisfies } \lim_{\vec{x} \rightarrow \vec{a}} \frac{f(\vec{x}) - h(\vec{x})}{\|\vec{x} - \vec{a}\|} = 0.$$

Practical tool

Theorem: Suppose $f: X \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ and $\vec{a}$ is an interior point of X .

If there is a neighbourhood Y of $\vec{a}$ on which $\vec{\nabla} f$ exists and is continuous, then f is differentiable at $\vec{a}$.

e.g. Show that $f(x, y, z) = x^2 y e^{z^2}$ is differentiable everywhere

$$\frac{\partial f}{\partial x}(x, y, z) = 2xy e^{z^2} = f_x(x, y, z)$$

$$\frac{\partial f}{\partial y}(x, y, z) = f_y(x, y, z) = x^2 e^{z^2}$$

$$\frac{\partial f}{\partial z}(x, y, z) = f_z(x, y, z) = 2z x^2 y e^{z^2}$$

Notice $\text{domain}(f) = \mathbb{R}^3$ and f is continuous on $\mathbb{R}^3$

and f_x, f_y, f_z are continuous on all $\mathbb{R}^3$

Therefore, f is differentiable on all $\mathbb{R}^3$

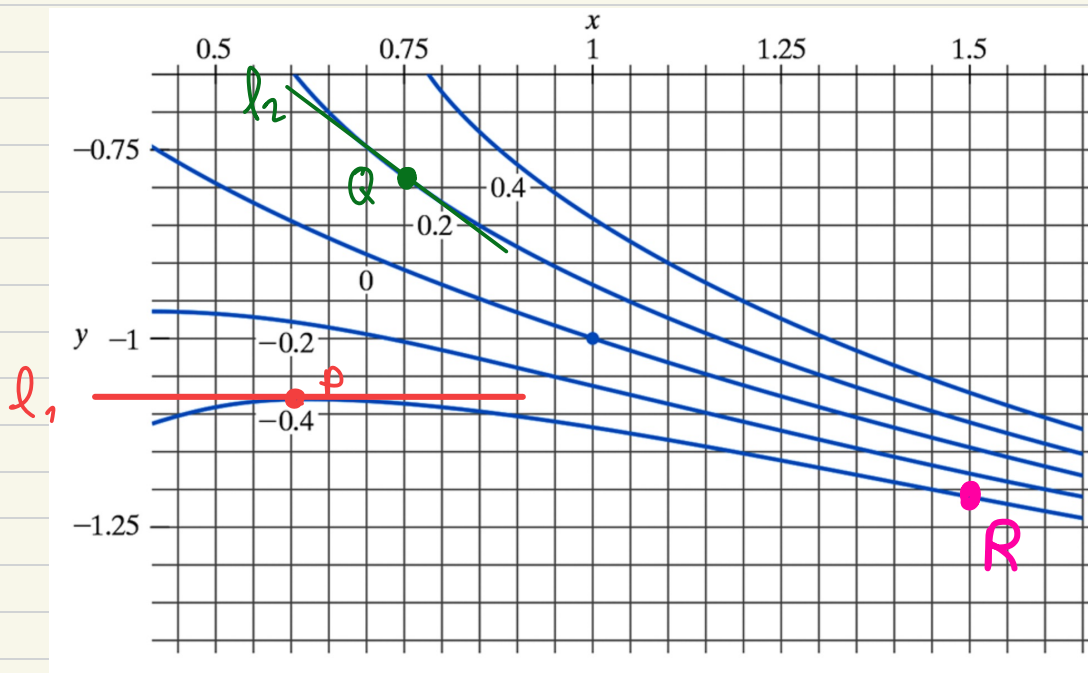
non-example: $f(x, y) = \begin{cases} 0 & , \text{ if } (x, y) = (0, 0) \\ \frac{xy}{x^2 + y^2} & , \text{ if } (x, y) \neq (0, 0) \end{cases}$

recall f is not continuous at $(0, 0) \Rightarrow f$ is not differentiable at $(0, 0)$

$$\text{but } \frac{\partial f}{\partial x}(0, 0) = 0 = \frac{\partial f}{\partial y}(0, 0)$$

def: We say that a function $f: \mathbb{X} \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$
 (with $\mathbb{X}$ open) is C^1 if all its partial
 derivatives exist and are continuous (on all $\mathbb{X}$)

Suppose this is the contour plot of a C^1 function



Is $f_x(P)$ bigger, equal or smaller than $f_x(Q)$

l_1 is tangent to the curve $f(x, y) = -0.4 \Rightarrow f_x(P) = 0$

l_2 is tangent to the curve $f(x, y) = 0.2$

What happens if you move along x-axis around Q? Positive movement
 gives increase in $f \Rightarrow f_x(Q) > 0$

$$f_y(P) > f_y(R) \\ \Leftrightarrow \Rightarrow f_y(P) < f_y(R)$$

Tangent plane (or hyperplane) to explicit graph of scalar function

Suppose $f: \Sigma \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}$ is differentiable at $\vec{a}$,
 then the tangent plane to the graph $z = f(x, y)$ is given by
 $z = h(x, y)$ where $h(x, y) = f(\vec{a}) + f_x(x - a_x) + f_y(y - a_y)$

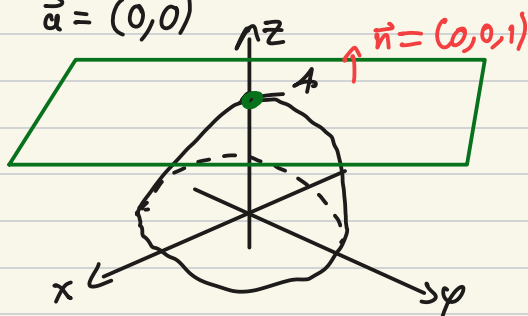
example from last class

• $f(x, y) = 4 - x^2 - y^2$ at $\vec{a} = (0, 0)$

$$\frac{\partial f}{\partial x}(x, y) = -2x$$

$$\frac{\partial f}{\partial y}(x, y) = -2y$$

f is diff everywhere



around $\vec{a}$ $f(\vec{x}) \approx h(x, y)$

$$h(x, y) = f(\vec{a}) + f_x(\vec{a})(x - a_x) + f_y(\vec{a})(y - a_y)$$

$$= 4 + 0 \cdot (x - 0) + 0 \cdot (y - 0) \leadsto z = 4$$

• $f(x, y) = 4x^2y + y^2$ at $\vec{a} = (1, 2)$

$$\frac{\partial f}{\partial x}(x, y) = 8xy$$

$$\frac{\partial f}{\partial y}(x, y) = 4x^2 + 2y$$

$$\vec{\nabla} f(\vec{a}) = (16, 8)$$

$$\vec{h}(x, y) = (16, 8) \cdot (x - 1, y - 2) + 12$$

plane: $z = 12 + 16(x - 1) + 8(y - 2)$

point of tangency in the graph $(1, 2, 12) = (\vec{a}, f(\vec{a}))$

recall: A point has dimension 0

A line (or in general a curve^{*}) has dimension 1

A plane (or in general a surface^{*}) has dimension 2

"def": The corresponding analogs are:

a hyperplane (or in general a hypersurface) with dimension n
(inside of $\mathbb{R}^{n+1}$)

* you can ignore this for now,
but for the sake of rigor
we need to add "without
self-intersection"

def: A hyperplane inside $\mathbb{R}^{n+1}$ is the set (with dimension n)
of solutions to a single linear equation (i.e. $\vec{m} \cdot \vec{r} = d$)

e.g. hyperplane in $\mathbb{R}^2$ is a line: $ax + by = d$

hyperplane in $\mathbb{R}^3$ is a plane: $ax + by + cz = d$

hyperplane in $\mathbb{R}^4$ has eq. $ax_1 + bx_2 + cx_3 + dx_4 = d$

Tangent hyperplane to a graph

e.g. $f(x, y, z) = x^2 + y^2 + z$ tangent hyperplane
to f at $(1, 1, 2)$

↳ $w = f(x, y, z)$ gives a hypersurface inside $\mathbb{R}^4$

$w = f(1, 1, 2) + \nabla f|_{(1,1,2)} \cdot (\vec{x} - \vec{a})$ gives a hyperplane
tangent to $w = f(x, y, z)$
at $\vec{a}$.

$$\nabla f(x, y, z) = (2x, 2y, 1)$$

$$\nabla f|_{(1,1,2)} = (2, 2, 1)$$

$$w = 4 + (2, 2, 1) \cdot (x-1, y-1, z-2)$$

$$w = f(\vec{a}) + \nabla f|_{\vec{a}} \cdot (\vec{x} - \vec{a})$$